COMP 2711
Discrete Math Tools for Computer Science
Mock Midterm Examination 01

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Use propositional laws, without a truth table, to simplify the following to a single literal:

$$\neg((p \vee q) \wedge (p \vee \neg q)).$$

(b) **[2 points]** Is the argument $p \to q,\ q\ \therefore p$ valid? Justify using the definition of validity.

(c) **[5 points]** From $p \vee q$, $\neg p \vee r$, $q \to s$, $r \to s$, and $s \to t$, derive t. Give a formal derivation naming the rules.

Problem 2: Predicate logic [13 points]

(a) **[4 points]** The domain is all people. $S(x)$ means student and $R(x, y)$ means x reviewed y's report. Translate: (i) every student reviewed some student's report; (ii) some student's report was reviewed by no student. Each translation is worth 2 points.

(b) **[4 points]** Negate $\forall x(A(x) \rightarrow \exists y(B(y) \wedge \forall z\, C(x, y, z)))$, placing every negation immediately before a predicate.

(c) **[5 points]** Over a nonempty domain, prove formally that $\forall x(P(x) \rightarrow Q(x))$, $\forall x(Q(x) \rightarrow \neg R(x))$, and $\exists x(P(x) \wedge S(x))$ imply $\exists x(S(x) \wedge \neg R(x))$.

Problem 3: Sets and functions [12 points]

Use $\mathbb{N} = \{0, 1, 2, \ldots\}$.

(a) **[2 points]** For sets A, B, C, prove $(A \setminus B) \cap C = A \cap (C \setminus B)$ by element membership.

(b) **[3 points]** Let $O = \{2k+1 : k \in \mathbb{Z}\}$ and $f : \mathbb{Z} \to O$, $f(k) = 2k+1$. Prove f is bijective and give its inverse with domain and codomain.

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) $\varnothing \in \{\varnothing\}$.

(ii) $\{\varnothing\} \subseteq \varnothing$.

(iii) $|\mathcal{P}(\{a, b\})| = 4$ when $a \neq b$.

(iv) $\{(1, a), (1, b), (2, a)\}$ is a relation from $\{1, 2\}$ to $\{a, b\}$.

(v) An injective function must be onto its stated codomain.

(vi) The set of all finite binary strings is countably infinite.

(vii) The set of all infinite binary sequences is countable.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** Three distinct tasks are placed in an ordered schedule, chosen without reuse from six distinct tasks. How many schedules include task T?

(b) **[2 points]** Let $U = \{1, 2, 3, 4\}$ and $V = \{a, b, c\}$. Count (i) functions $f : U \to V$ with $f(1) = f(2)$; (ii) injective functions $U \to V$.

(c) **[2 points]** Encode $S = \{1, 3, 5\} \subseteq \{1, 2, 3, 4, 5\}$ by its membership bit string in label order. Use this encoding to count all subsets of $\{1, 2, 3, 4, 5\}$.

(d) **[2 points]** The standard triple-loop algorithm computes AB for a 3×4 matrix A and a 4×5 matrix B, using four scalar multiplications per output entry. How many scalar multiplications occur?

(e) **[2 points]** A code has length two or three over $\{A, B, C, 0, 1\}$, begins with a letter, and contains at least one digit. Repetition is allowed. Count the codes.

(f) **[2 points]** Arrange the seven distinct letters A, B, C, D, E, F, G in a row, with A, B, C together in any internal order. Count the arrangements.

(g) **[2 points]** Six distinct people sit around a circular table with A and B adjacent. Rotations are equivalent; reflections are different. Count the seatings.

(h) **[2 points]** Find the coefficient of x^4 in $(2x-1)^7$.

Problem 5: Combinatorial proof [10 points]

[**10 points**] For $2 \leq r \leq n$, prove combinatorially that

$$r(r-1)\binom{n}{r} = n(n-1)\binom{n-2}{r-2}.$$

Problem 6: Inclusion-exclusion [9 points]

[9 points] Use inclusion–exclusion to count integer solutions of $x+y+z = 12$ with $1 \leq x \leq 5$, $0 \leq y \leq 4$, and $2 \leq z \leq 7$.

Problem 7: Pigeonhole principle [7 points]

[**7 points**] Any sixteen distinct integers selected from $\{1, 2, \ldots, 30\}$ include a pair with sum 31 and a pair with difference 15. The pairs may differ. Prove both claims and show that sixteen is the smallest number guaranteeing both.

Problem 8: Sampling and events [6 points]

Six distinct tokens are labelled $R_1, R_2, R_3, B_1, B_2, G$. Draw three in order without replacement; every ordered triple of distinct tokens is equally likely. Let E and F mean that the first and second tokens, respectively, are red.

(a) **[1 points]** Describe the sample space and its size.

(b) **[1 points]** Find $P((R_1, B_1, G))$.

(c) **[1 points]** Find $P(E \cap F)$.

(d) **[1 points]** Find the probability of no red token.

(e) **[2 points]** Are E, F independent? Are they disjoint? Justify both.

Problem 9: A noisy inspection [7 points]

A randomly selected component is defective with probability 1/10. An alarm sounds with probability 4/5 for a defective component and 1/5 for a good one. The outcomes are the defect status and alarm status with these conditional probabilities.

(a) **[3 points]** Find the probability that the alarm sounds.

(b) **[4 points]** Find the probability of a defect given an alarm, and given no alarm.

Problem 10: Moments and repeated measurements [10 points]

D takes $0, 1, 2$ with probabilities $1/4, 1/2, 1/4$. All copies of D mentioned below are mutually independent and have this law.

(a) **[2 points]** Find $E(D)$ and $V(D)$.

(b) **[2 points]** For $Y = 3 - 2D$, find $E(Y)$ and $V(Y)$.

(c) **[2 points]** For an independent copy D', compare $E(DD')$ with $E(D^2)$.

(d) **[4 points]** Let $M = (D_1 + D_2 + D_3)/3$. Find its mean, variance and standard deviation, explaining the role of independence.

Draft Paper

Mock examination working space.

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